INCH-POUND
MIL-PRF-3098/77F
7 January 2010
SUPERSEDING
MIL-PRF-3098/77E
12 January 2005

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR5/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2 MHz to 10 MHz; fundamental; noncontrolled; antiresonance.

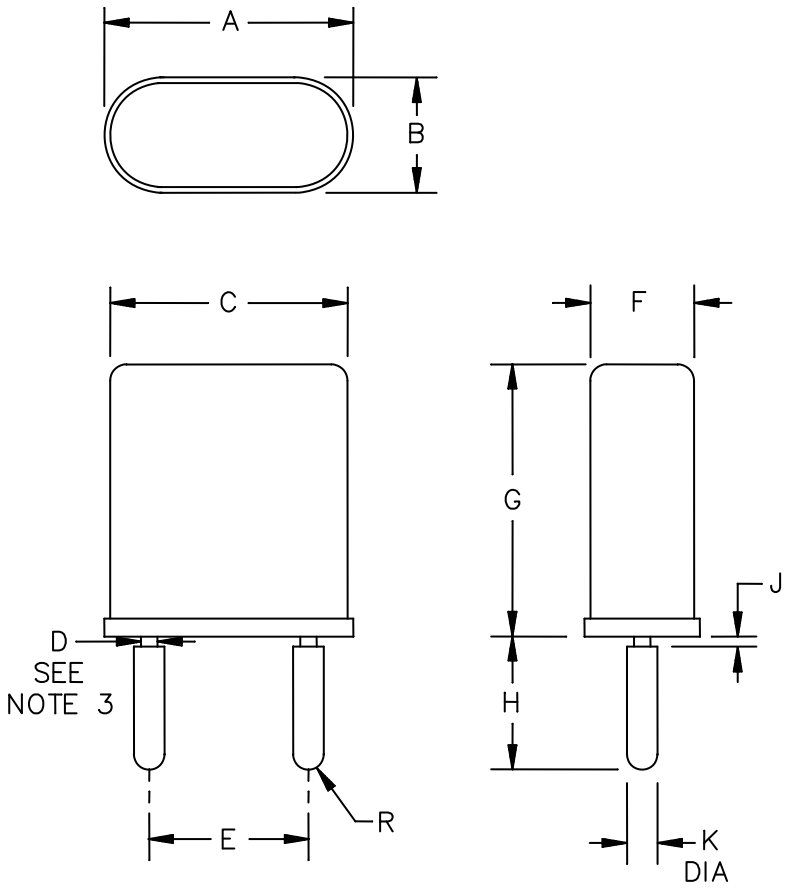

FIGURE 1. Crystal unit - CR5/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.757	---	19.23
B	---	.352	---	8.94
C	---	.752	---	19.10
D	.048	.052	1.22	1.32
E	.478	.494	12.14	12.55
F	---	.317	---	8.05
G	---	.775	---	19.68
H	---	.448	---	11.38
J	.012	.047	0.30	1.19
K	.091	.095	2.31	2.41

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR5/U - Continued.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.0 MHz to 10.0 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW maximum.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

Antiresonance, load capacitance: 25.0 pF ±0.5 pF.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

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Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
2.00 to 2.10	550
2.10+ to 2.225	500
2.225+ to 2.30	450
2.30+ to 2.45	400
2.45+ to 2.60	350
2.60+ to 2.80	300
2.80+ to 3.05	250
3.05+ to 3.40	200
3.40+ to 3.60	180
3.60+ to 3.90	150
3.90+ to 4.25	130
4.25+ to 4.65	100
4.65+ to 5.10	75
5.10+ to 6.50	50
6.50+ to 7.50	37
7.50+ to 9.00	25
9.00+ to 10.00	20

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Part or Identifying Number (PIN): CR5/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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Custodians:
Army - CR
Navy - EC
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DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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